

ESE5180: Lab 0 Zephyr

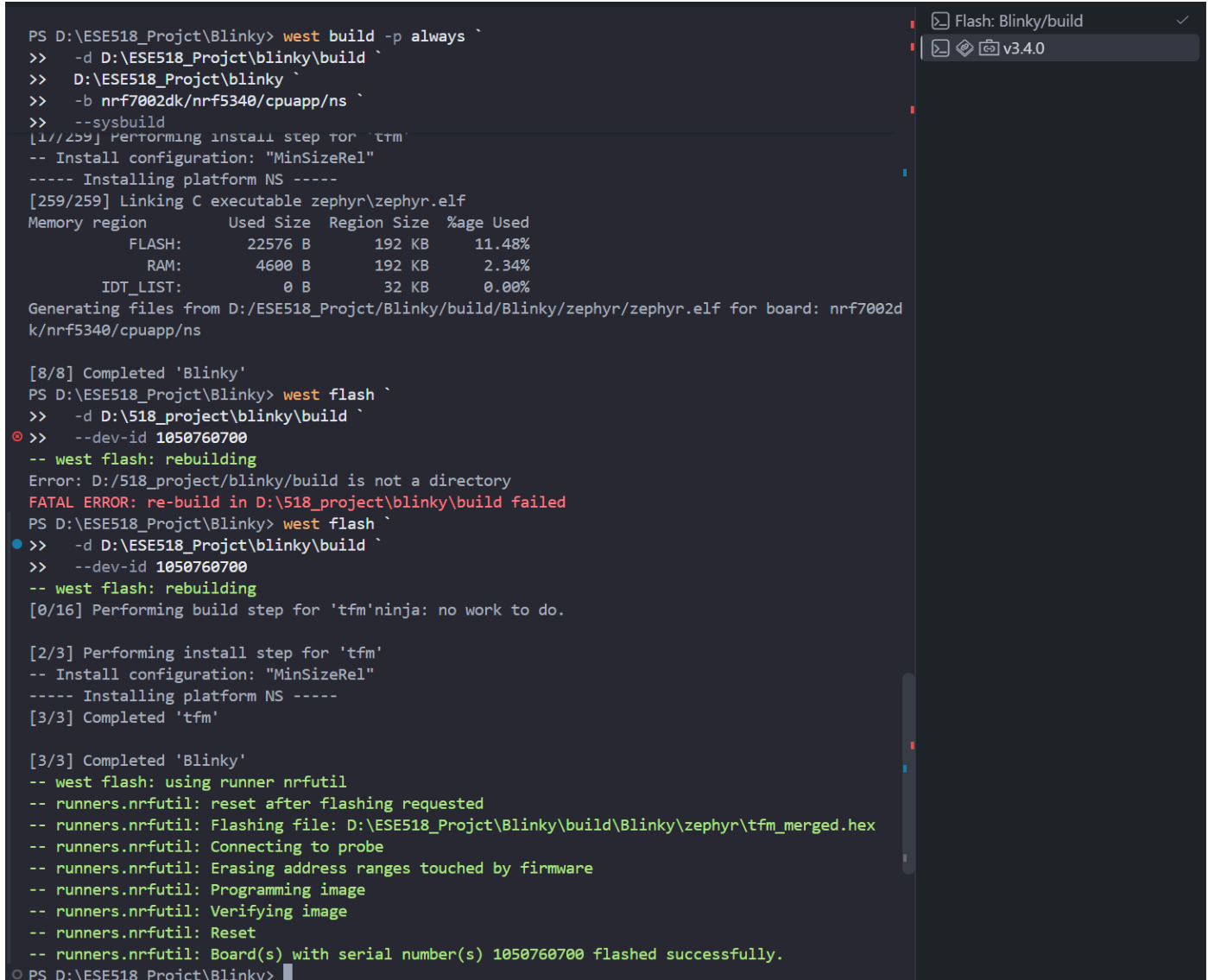
Team Member Name Email Address

Zicong Zhang zhang89@seas.upenn.edu

GitHub Repository URL: github.com/zhang89-spec/ESE5180-lab0

3. Building with West

3.1 terminal prints



```
PS D:\ESE518_Project\Blinky> west build -p always `
>> -d D:\ESE518_Project\blinky\build `
>> D:\ESE518_Project\blinky `
>> -b nrf7002dk/nrf5340/cpuapp/ns `
>> --sysbuild
[17/259] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[259/259] Linking C executable zephyr\zephyr.elf
Memory region      Used Size  Region Size  %age Used
FLASH:             22576 B      192 KB       11.48%
RAM:               4600 B       192 KB        2.34%
IDT_LIST:           0 B         32 KB        0.00%
Generating files from D:\ESE518_Project\Blinky\build\Blinky\zephyr\zephyr.elf for board: nrf7002d
k/nrf5340/cpuapp/ns

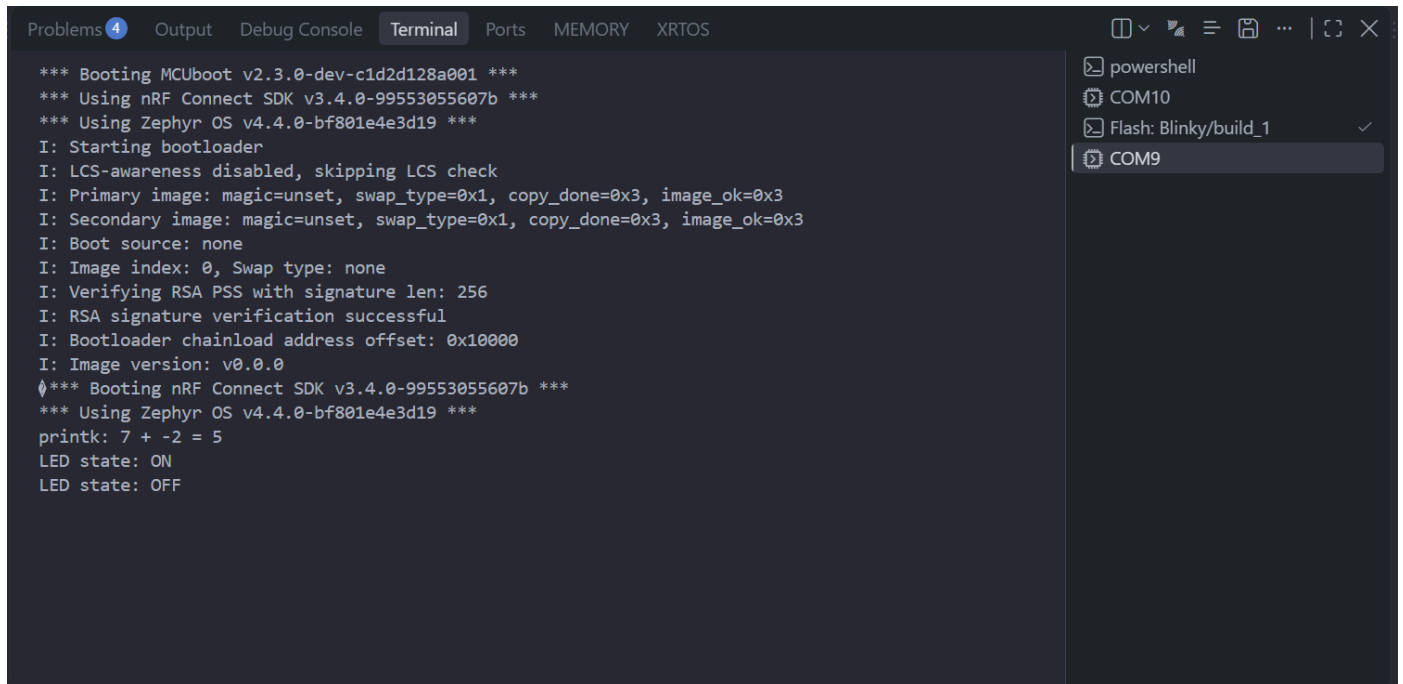
[8/8] Completed 'Blinky'
PS D:\ESE518_Project\Blinky> west flash `
>> -d D:\518_project\blinky\build `
>> --dev-id 1050760700
-- west flash: rebuilding
Error: D:\518_project\blinky\build is not a directory
FATAL ERROR: re-build in D:\518_project\blinky\build failed
PS D:\ESE518_Project\Blinky> west flash `
>> -d D:\ESE518_Project\blinky\build `
>> --dev-id 1050760700
-- west flash: rebuilding
[0/16] Performing build step for 'tfm'ninja: no work to do.

[2/3] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[3/3] Completed 'tfm'

[3/3] Completed 'Blinky'
-- west flash: using runner nrfutil
-- runners.nrfutil: reset after flashing requested
-- runners.nrfutil: Flashing file: D:\ESE518_Project\Blinky\build\Blinky\zephyr\tfm_merged.hex
-- runners.nrfutil: Connecting to probe
-- runners.nrfutil: Erasing address ranges touched by firmware
-- runners.nrfutil: Programming image
-- runners.nrfutil: Verifying image
-- runners.nrfutil: Reset
-- runners.nrfutil: Board(s) with serial number(s) 1050760700 flashed successfully.
PS D:\ESE518_Project\Blinky>
```

6. Printing vs. Logging

6.1 Console output of printk

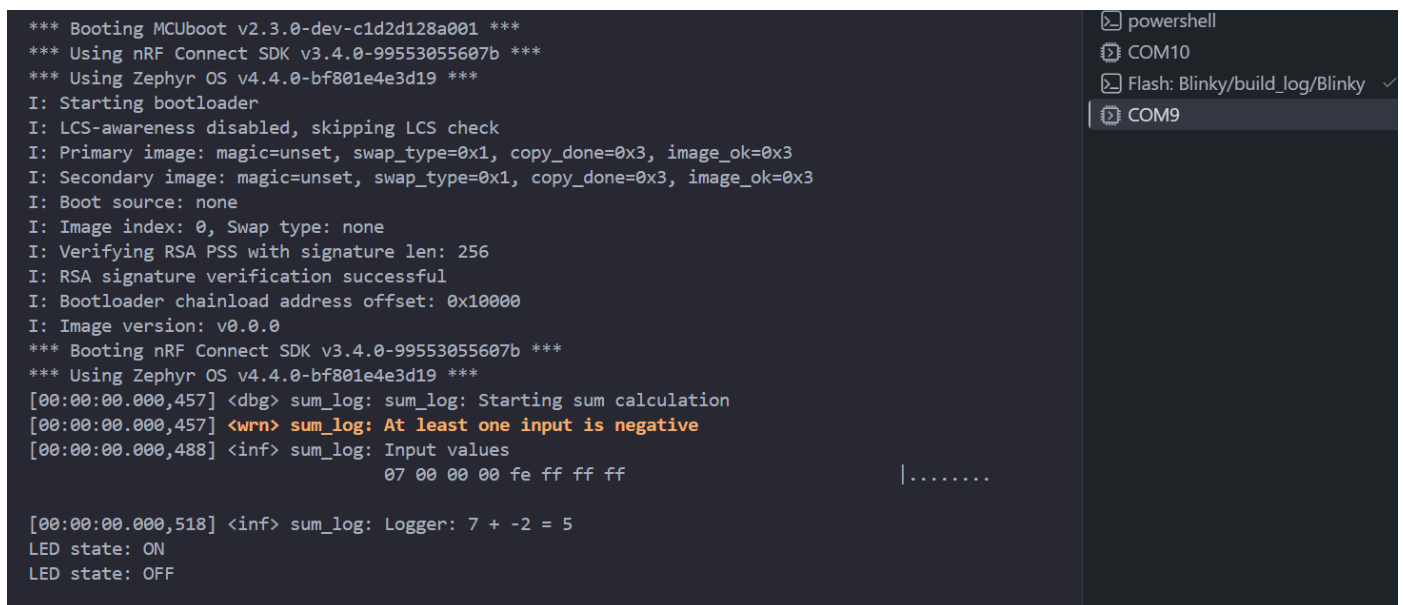


The screenshot shows a terminal window with the following content:

```
*** Booting MCUboot v2.3.0-dev-c1d2d128a001 ***
*** Using nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
I: Starting bootloader
I: LCS-awareness disabled, skipping LCS check
I: Primary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Secondary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Boot source: none
I: Image index: 0, Swap type: none
I: Verifying RSA PSS with signature len: 256
I: RSA signature verification successful
I: Bootloader chainload address offset: 0x10000
I: Image version: v0.0.0
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
printk: 7 + -2 = 5
LED state: ON
LED state: OFF
```

On the right side, there is a sidebar with a list of devices: powershell, COM10, Flash: Blinky/build_1, and COM9. COM9 is selected and highlighted.

6.2 Console output of log



The screenshot shows a terminal window with the following content:

```
*** Booting MCUboot v2.3.0-dev-c1d2d128a001 ***
*** Using nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
I: Starting bootloader
I: LCS-awareness disabled, skipping LCS check
I: Primary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Secondary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Boot source: none
I: Image index: 0, Swap type: none
I: Verifying RSA PSS with signature len: 256
I: RSA signature verification successful
I: Bootloader chainload address offset: 0x10000
I: Image version: v0.0.0
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
[00:00:00.000,457] <dbg> sum_log: sum_log: Starting sum calculation
[00:00:00.000,457] <wrn> sum_log: At least one input is negative
[00:00:00.000,488] <inf> sum_log: Input values
                        07 00 00 00 fe ff ff ff
                        |.....

[00:00:00.000,518] <inf> sum_log: Logger: 7 + -2 = 5
LED state: ON
LED state: OFF
```

On the right side, there is a sidebar with a list of devices: powershell, COM10, Flash: Blinky/build_log/Blinky, and COM9. COM9 is selected and highlighted.

7. Ztest for Unit Testing

7.1 testing outputs

```

PS D:\ESE518_Projct\lab0-zephyr-skeleton\Blinky\tests\SUM_UNIT_TEST> west build -d build -t run
[0/1] To exit from QEMU enter: CTRL+X [QEMU] CPU: cortex-m3c.\Program Files\qemu\qemu-system-arm.exe: -mon chardev=con,mode=readline: warning: '-mon' is deprecated, use '-object' with 'monitor-hmp' or 'monitor-qmp' types instead
C:\Program Files\qemu\qemu-system-arm.exe: warning: nic stellaris_enet.0 has no peer
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
Running TESTSUITE sum_log_test_suite
=====
START - test_sum_log_basic
D: Starting sum calculation
I: Input values
I: 02 00 00 00 03 00 00 00 |.....
I: Logger: 2 + 3 = 5

Assertion succeeded at CMAKE_SOURCE_DIR/src/sum_test.c:9 (sum_log_test_suite_test_sum_log_basic)
PASS - test_sum_log_basic in 0.002 seconds
=====
START - test_sum_log_negative
D: Starting sum calculation
W: At least one input is negative
I: Input values
I: fb ff ff ff 02 00 00 00 |.....
I: Logger: -5 + 2 = -3

Assertion succeeded at CMAKE_SOURCE_DIR/src/sum_test.c:18 (sum_log_test_suite_test_sum_log_negative)
PASS - test_sum_log_negative in 0.002 seconds
=====
START - test_sum_log_zero
D: Starting sum calculation
I: Input values
I: 00 00 00 00 00 00 00 00 |.....
I: Logger: 0 + 0 = 0

Assertion succeeded at CMAKE_SOURCE_DIR/src/sum_test.c:27 (sum_log_test_suite_test_sum_log_zero)
PASS - test_sum_log_zero in 0.002 seconds
=====
TESTSUITE sum_log_test_suite succeeded

----- TESTSUITE SUMMARY START -----

SUITE PASS - 100.00% [sum_log_test_suite]: pass = 3, fail = 0, skip = 0, total = 3 duration = 0.006
seconds
- PASS - [sum_log_test_suite.test_sum_log_basic] duration = 0.002 seconds
- PASS - [sum_log_test_suite.test_sum_log_negative] duration = 0.002 seconds
- PASS - [sum_log_test_suite.test_sum_log_zero] duration = 0.002 seconds

----- TESTSUITE SUMMARY END -----

=====

```

8. Adding a Peripheral (BME280)

8.1 Temperature output

```
*** Booting MCUboot v2.3.0-dev-c1d2d128a001 ***
*** Using nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
I: Starting bootloader
I: LCS-awareness disabled, skipping LCS check
I: Primary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Secondary image: magic=unset, swap_type=0x1, copy_done=0x3, image_ok=0x3
I: Boot source: none
I: Image index: 0, Swap type: none
I: Verifying RSA PSS with signature len: 256
I: RSA signature verification successful
I: Bootloader chainload address offset: 0x10000
I: Image version: v0.0.0
❖*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
printk: 7 + -2 = 5
[00:00:00.022,766] <inf> bme280_raw: BME280 detected at I2C address 0x77
[00:00:00.023,620] <inf> blinky_app: Temperature: 26.14 C
[00:00:02.034,057] <inf> blinky_app: Temperature: 26.15 C
[00:00:04.044,494] <inf> blinky_app: Temperature: 26.15 C
[00:00:06.054,962] <inf> blinky_app: Temperature: 26.16 C
```

8.2 Ztest Output

```
PS D:\ESE518_Projct\lab0-zephyr-skeleton\Blinky\tests\BME280_UNIT_TEST> west build -d build -t run
Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:19 (bme280_test_suite_test_devicetree_
setup)

Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:21 (bme280_test_suite_test_devicetree_
setup)
PASS - test_devicetree_setup in 0.001 seconds
=====
START - test_invalid_arguments

Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:54 (bme280_test_suite_test_invalid_arg
uments)

Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:57 (bme280_test_suite_test_invalid_arg
uments)

Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:60 (bme280_test_suite_test_invalid_arg
uments)
PASS - test_invalid_arguments in 0.002 seconds
=====
START - test_temperature_compensation

Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:38 (bme280_test_suite_test_temperature_
compensation)

Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:39 (bme280_test_suite_test_temperature_
compensation)

Assertion succeeded at CMAKE_SOURCE_DIR/src/bme280_test.c:41 (bme280_test_suite_test_temperature_
compensation)
PASS - test_temperature_compensation in 0.002 seconds
=====
TESTSUITE bme280_test_suite succeeded

----- TESTSUITE SUMMARY START -----

SUITE PASS - 100.00% [bme280_test_suite]: pass = 3, fail = 0, skip = 0, total = 3 duration = 0.005
seconds
- PASS - [bme280_test_suite.test_devicetree_setup] duration = 0.001 seconds
- PASS - [bme280_test_suite.test_invalid_arguments] duration = 0.002 seconds
- PASS - [bme280_test_suite.test_temperature_compensation] duration = 0.002 seconds

----- TESTSUITE SUMMARY END -----

=====
PROJECT EXECUTION SUCCESSFUL
```